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Explainable vision transformer for coffee leaf nutrient deficiency detection with rule-based treatment recommendation

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Abstract

Coffee is a worldwide significant agricultural commodity, but nutritional deficits in coffee leaves impair plant health and output, necessitating early and correct diagnosis. Existing deep learning algorithms confront limitations such as common deficiency symptoms, limited datasets, and poor interpretability. This study presents an Explainable Vision Transformer (XViT)-based system for detecting coffee leaf nutrient deficiencies and making rule-based treatment recommendations. The approach improves data variety by preprocessing and class-wise augmentation, whereas XViT collects global and detailed visual patterns for classification. An expert-defined rule-based module makes suitable treatment recommendations, while Grad-CAM visualization enhances model clarity by emphasizing key symptom locations. Experimental results show that the proposed XViT model regularly outperforms baseline architectures such as ResNet-50, EfficientNet-B4, and ConvNeXt. On the initial dataset, XViT achieved 94% precision, 91% recall, 92% F1-score, and accuracy of $94 \pm 0.37\%$, whereas augmentation improved accuracy to $96 \pm 0.29\%$. The rule-based recommendation module achieved 100% rule coverage, 100% rule consistency, a 93.72% end-to-end recommendation success rate, and a Cohen's Kappa agreement of 0.93 among agricultural specialists. Grad-CAM study indicated that the model concentrates on significant symptom regions linked with nutrient deficits, hence increasing transparency and reliability. Overall, the proposed framework for precision coffee nutrient management is accurate, explainable, and practical since it combines deep learning-based diagnosis with expert-guided treatment.

Keywords Vision transformer, Nutrient deficiency detection, Explainable artificial intelligence, Coffee leaf management, Precision agriculture, Rule-based recommendation

1 Introduction

Coffee is one of the world's most economically important crops, providing a significant income for millions of growers in tropical and subtropical countries. However, nutritional deficits significantly impact coffee production by reducing plant health, crop yield, and bean quality [1, 2]. Nutrient deficiencies in nitrogen, phosphorus, potassium,



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magnesium, and iron are common apparent signs on coffee leaves. If not recognized promptly, these flaws can dramatically affect output and overall coffee quality [2, 3].

Traditional approaches for diagnosing nutrient deficiencies mainly rely on manual inspection, soil testing, and laboratory analysis [3, 4]. Although these methods are frequently used, they are time-consuming, expensive, and require specialist knowledge. Furthermore, manual monitoring is difficult to implement successfully in large-scale coffee plantations, which limits its scalability and effectiveness [5, 6]. These limitations underscore the importance of automated, reliable nutrient-deficit detection systems for precision agriculture.

Recent advancements in artificial intelligence (AI) and deep learning have greatly enhanced automated plant health monitoring. In particular, convolutional neural networks (CNNs) have demonstrated promising performance in agricultural image analysis applications [7, 8]. However, CNN-based approaches frequently fail to capture global contextual information and subtle visual patterns for nutrient-deficit detection in complex agricultural imagery [9]. Vision Transformers (ViTs) have recently emerged as an effective alternative due to their self-attention mechanisms and ability to model long-range interactions in images [10–12]. These properties make transformer-based techniques useful for detecting nutritional deficits in coffee leaves [13–16].

Despite recent progress, existing studies still have numerous limitations. The majority of past research has focused on detecting plant diseases rather than identifying nutrient deficiencies. Furthermore, many current systems only provide categorization results, not treatment recommendations or interpretability analyses. Furthermore, transformer-based learning techniques for detecting coffee nutrient deficiencies are underexplored.

To address these issues, this study provides an Explainable Vision Transformer (ViT) framework for automated nutrient deficiency identification and rule-based treatment recommendations in coffee leaves. The proposed architecture uses a Vision Transformer to gather global contextual information as well as subtle visual patterns in order to accurately classify nutrient deficiencies. Based on the expected nutrient deficiency, an expert-defined rule-based suggestion module automatically applies the appropriate agronomic treatment in accordance with accepted nutrition-treatment criteria. Grad-CAM is used to depict the discriminative regions that contribute to the model's predictions, which improves transparency and helps with practical decision-making. The proposed framework intends to support precision agriculture by offering an accurate, scalable, interpretable, and practical decision-support tool for managing coffee nutritional deficiencies.

The main contributions of this study are as follows:

1. An explainable vision transformer (ViT) framework is developed for automated detection of multiple nutrient deficiencies in coffee leaves by effectively capturing global contextual features and subtle visual symptoms.
2. An expert-defined rule-based treatment recommendation module is integrated to automatically generate appropriate agronomic treatment recommendations based on the predicted nutrient deficiency, enabling an end-to-end decision-support workflow.
3. A Grad-CAM-based explainable AI module is incorporated to improve model transparency by highlighting the discriminative regions of coffee leaf images that influence nutrient deficiency predictions.

4. Extensive experiments and comparative evaluations against state-of-the-art baseline models are conducted to demonstrate the effectiveness, robustness, and generalization capability of the proposed framework.
5. The proposed framework provides an accurate, interpretable, scalable, and practical decision-support solution for precision coffee agriculture by combining reliable nutrient deficiency detection with expert-guided treatment recommendations.

1.1 Literature review

Automated plant disease diagnosis systems have been greatly enhanced by recent developments in computer vision, machine learning, and deep learning. Image-based disease detection techniques have drawn a lot of interest in coffee farming because they can facilitate early diagnosis, lower the cost of manual inspections, and increase crop management effectiveness. Convolutional neural networks (CNNs), transfer learning techniques, hybrid deep learning architectures, and, more recently, transformer-based models are the main methods used in current research to classify coffee leaf diseases.

CNN-based architectures were widely used in early research on coffee leaf disease detection due to their powerful capacity to extract discriminative visual information from leaf images. Under controlled imaging settings, CNN models have been shown in multiple studies to accurately discern between healthy and sick coffee leaves. For example, utilizing over 4000 images of coffee leaves, a CNN-based framework for identifying coffee wilt disease achieved a classification accuracy of 97.9%, demonstrating the efficacy of deep feature extraction for disease recognition [17]. Similar to this, CNN models based on transfer learning, like MobileNet and ResNet50, were used to classify coffee rust, brown eye spot, wilt, and healthy leaves with an accuracy of more than 97% [18]. The effectiveness of CNN architectures in collecting local texture and color differences linked to coffee leaf diseases was validated by this research.

Many studies used transfer learning approaches with pretrained deep learning models to further enhance classification performance and solve the problem of limited labeled datasets. In coffee illness classification tasks, pretrained architectures such as ResNet50, InceptionV3, VGG16, Xception, and ResNet101 have demonstrated exceptional performance [19, 20]. In fact, research on large-scale coffee leaf datasets reported classification accuracies of up to 99%, confirming transfer learning's robustness for agricultural image analysis. These techniques' success is largely due to their capacity to apply feature representations learnt from huge benchmark datasets to domain-specific agricultural tasks. Despite their excellent classification accuracy, most extant transfer learning studies are exclusively concerned with disease identification and rarely address nutrient deficiency detection or treatment recommendations.

Beyond deep CNN structures, classic machine learning and ensemble learning techniques have been investigated for coffee crop monitoring applications. Ensemble-based techniques, such as XGBoost, have been employed to detect Coffee Berry Disease early on, producing competitive results with accuracies of 98% [21]. These methods demonstrated that handmade feature extraction mixed with ensemble classifiers can still deliver dependable performance in specific agricultural contexts. Nonetheless, traditional machine learning approaches can necessitate extensive feature engineering and may fail to generalize effectively under changing environmental conditions when compared to deep learning models.

Recent research has increasingly focused on hybrid and advanced deep learning architectures to improve coffee leaf disease detection. Hybrid models that incorporate architectures such as U-Net, SegNet, MobileNetV3, VGG16, and Swin Transformers have shown improved feature representation and classification performance [22, 23]. Transformer-based models, in particular, have received interest because of their self-attention processes and capacity to capture global contextual information, making them useful for detecting subtle illness indications in agricultural images.

Advanced techniques, such as Graph Neural Networks (GNNs), have also been used to monitor agricultural activities. Hybrid Vision Graph Neural Networks (HV-GNNs) excelled at detecting coffee pests and crop damage by accurately modeling spatial relationships in complex agricultural situations [24].

Beyond coffee crops, deep learning algorithms have been widely used to detect disease and nutrient deficiencies in crops such as rice, maize, wheat, and tomato [25]. Transformer-based models have recently demonstrated promising results in plant phenotyping and nutrient deficiency diagnosis, owing to their capacity to capture subtle visual fluctuations and long-range correlations within images [26].

Although previous research has yielded good classification accuracy for coffee leaf disease identification, a number of significant research gaps remain. First, most present research focuses on disease categorization rather than nutritional deficit identification, despite the fact that nutrient deficiencies have a major impact on coffee growth, production, and quality. Second, many existing techniques conclude with nutrient deficiency classification but do not provide actionable treatment recommendations to help farmers make timely management decisions. Third, traditional convolutional neural network (CNN) models frequently fail to capture the global contextual relationships and subtle visual patterns linked to nutrient insufficiency symptoms.

Furthermore, many contemporary systems do not use explainable artificial intelligence (XAI) methodologies, which limits the transparency, interpretability, and practical application of their predictions in real-world agricultural contexts. Although Vision Transformers have shown high performance in a variety of computer vision tasks, their application to explainable coffee leaf nutrient deficit detection combined with expert-guided treatment advice is mostly unexplored. Motivated by these constraints, this study provides an Explainable Vision Transformer (ViT) framework for detecting coffee leaf nutrient deficiencies and recommending rule-based treatments. The proposed system uses Vision Transformers' global feature representation capacity to accurately detect nutrient shortages in coffee leaves. Based on the expected shortfall, an expert-defined rule-based suggestion module creates the appropriate agronomic solution using known nutrient management principles. In addition, a Grad-CAM-based explainability module is included to depict the image regions that contribute the most to the model's predictions, increasing transparency, interpretability, and user trust. The proposed framework for precision coffee agriculture combines accurate nutrient shortage diagnosis, expert-guided treatment recommendations, and explainable artificial intelligence to create a practical and dependable decision-support system.

2 Methodology

2.1 Overview of the proposed framework

The proposed approach combines an Explainable Vision Transformer (XViT) for automated nutrient deficit detection and an expert-defined rule-based module for treatment recommendations in coffee leaves. The framework functions as an end-to-end decision-support system, first analyzing coffee leaf images to diagnose nutrient deficiencies, and then converting the projected deficiency categories into appropriate treatment recommendations based on specified agronomic principles. The input images are pre-processed and separated into fixed-size patches (16×16). These patches are embedded with positional encodings before being processed through several transformer encoder layers using multi-head self-attention methods. The obtained transformer representations allow for the efficient extraction of global contextual information and delicate visual patterns associated with nutrient insufficiency symptoms. Furthermore, a Grad-CAM-based explainability module identifies the leaf regions that provide the greatest contribution to model predictions, boosting transparency and interpretation. By combining precise diagnosis, expert-guided recommendations, and visual explanations, the proposed framework offers a realistic solution for smart agriculture and precision coffee nutrition management. For model development and evaluation, the proposed framework was assessed using a stratified five-fold cross-validation strategy. In each fold, approximately 861 images (72% of the complete dataset) were used for training, 96 images (8% of the complete dataset; 10% of the training portion) were allocated for internal validation, and 239 images (20% of the complete dataset) were reserved as an independent test set to ensure unbiased performance evaluation. Figure 1 depicts the overall design of the proposed framework.

2.2 Dataset collection and preparation

The dataset used for this study consists of coffee leaf images collected from publicly available sources and manually organized into nutrient deficiency categories. The final dataset contains eight nutrient deficiency classes and one healthy class. Each deficiency class includes a corresponding treatment suggestion label, enabling the proposed Vision

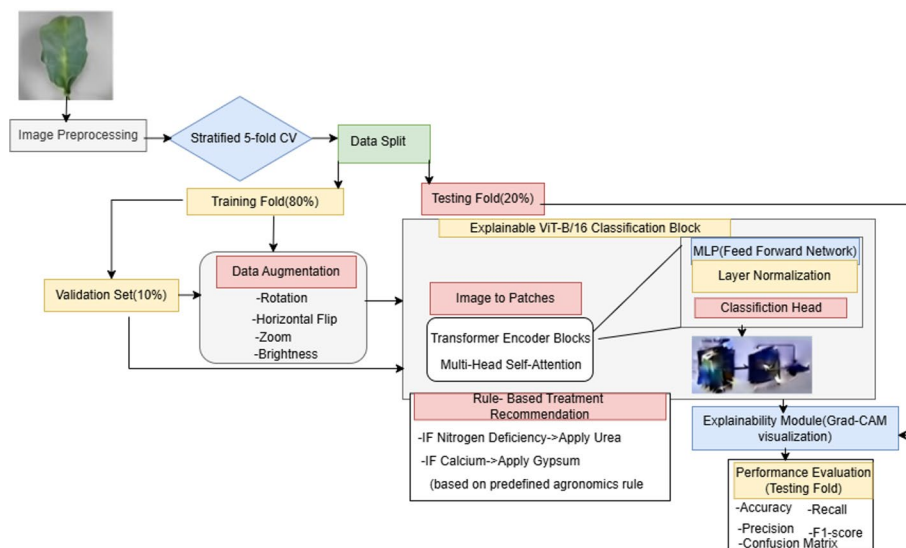


Fig. 1 The proposed architecture

Transformer to detect the leaf deficiency type and appropriate rule based treatment action.

2.2.1 Dataset collection

The main sources of photos of coffee leaves were two publicly accessible datasets. This study utilizes the CoLeaf dataset, a publicly available leaf image dataset designed for plant disease and nutrient deficiency analysis. In addition to the CoLeaf dataset, this study also employs the Plant Disease Classification Merged Dataset, a publicly available collection of leaf images compiled from multiple plant disease datasets. This merged dataset provides diverse disease categories and variations in illumination and background, which enhance the robustness of the proposed model. Images with unclear leaf conditions or several signs of nutritional stress were not included. As can be seen below, the dataset was finally divided into eight classes for nutrient deficiencies and a class for health in Table 1.

After merging data sources, the total number of samples became 1196 images, which were shuffled and split using a 5-fold cross-validation protocol.

2.2.2 Dataset preparation

First, every image was scaled to the 224×224 pixel fixed Vision Transformer input resolution. For uniform patch extraction and stable positional encoding in the ViT architecture, this resizing step guarantees that all input samples have the same spatial dimensions. A two-stage normalization procedure was then used to normalize the pixel values. First, each pixel's intensity was divided by 255 to scale it to the interval $[0,1]$. Second, to standardize feature distributions and speed up training convergence, standard ViT normalization based on the ImageNet mean (μ) and standard deviation (σ) was performed per RGB channel.

The sample images corresponding to each nutrient deficiency class and healthy leaves are presented in Fig. 2 below, providing a representative overview of the dataset.

Advanced online data augmentation approaches were used to boost robustness to real-world field variability and improve generalization performance. Random cropping, horizontal and vertical flips, and random rotations between 0° and 30° are examples of geometric transformations that mimic the variety of leaf orientation. In order to promote spatial invariance and enhance feature-learning under partial occlusion, local regions were masked or mixed using region-based perturbations like Cutout and

Table 1 The collected dataset description

Nutrient deficiency	Treatment recommendation	No. of images
Boron (B)	Borax (sodium tetraborate)	101
Calcium (Ca)	Gypsum (calcium sulfate)	162
Healthy	No treatment required	300
Iron (Fe)	Iron sulfate (FeSO_4)	65
Potassium (K)	Potassium sulfate (K_2SO_4)	96
Magnesium (Mg)	Magnesium sulfate (epsom salt)	79
Manganese (Mn)	Manganese sulfate (MnSO_4)	83
Nitrogen (N)	Urea	64
Phosphorus (P)	Triple superphosphate (TSP)	246
Total		1196

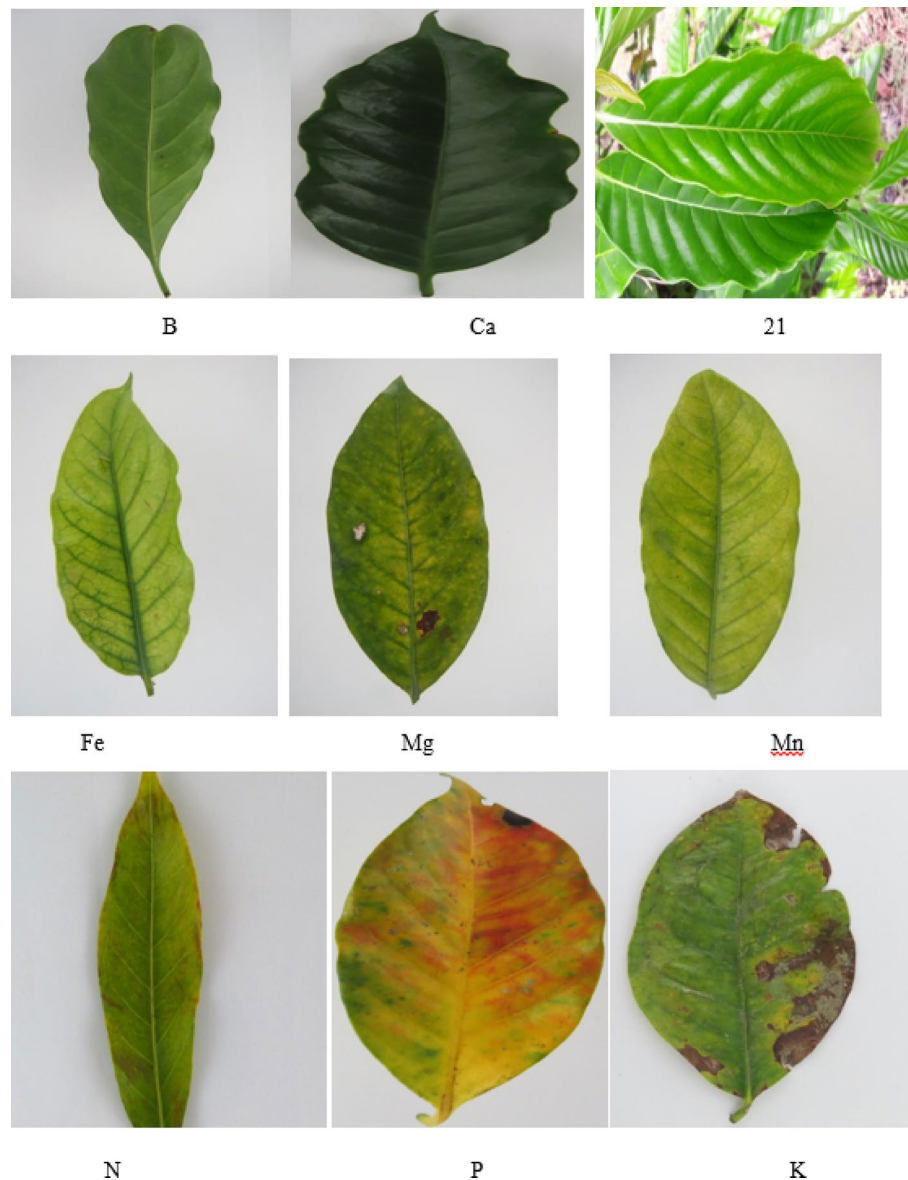


Fig. 2 Sample collected images

CutMix. Additionally, to provide smoother decision boundaries, labels and images were linearly blended at the pixel level using Mixup augmentation.

Brightness, contrast, saturation, and hue fluctuations are examples of photometric augmentations that were used to adjust for variations in illumination during field image acquisition. Together, these augmentation techniques add artificial diversity to training samples, which strengthens the model's resistance to changes in lighting, camera noise, leaf occlusion, and overlapping stress symptoms. This enhances the model's capacity for generalization when it is deployed in actual agricultural settings. The sample output images for augmented images are depicted in Fig. 3.

2.2.3 Annotation procedure and validation

Images from the CoLeaf Collection and Plant Disease Classification. Merged collections were evaluated and annotated to ensure they met the study's objectives. Each image was

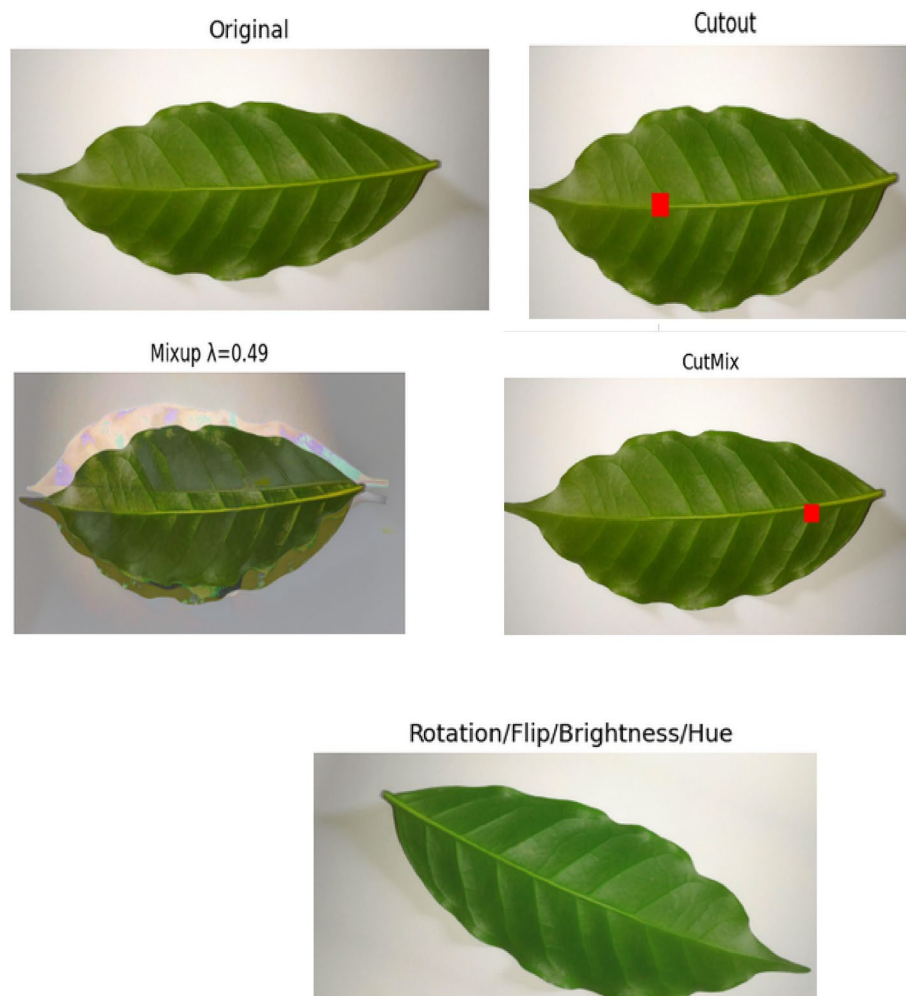


Fig. 3 Augmented sample images

classified as nutrient deficit or healthy based on dominant visual symptoms linked with distinct nutritional states of coffee plants.

The annotation method was guided by established agronomic knowledge of coffee plant nutrient shortages, which included visible traits such as chlorosis (yellowing), necrosis (tissue death), interveinal discoloration, and leaf deformation patterns. Images with confusing symptoms, multiple overlapping nutrient shortages, non-coffee leaf content, or low visual quality were eliminated in order to reduce annotation ambiguity and increase dataset reliability.

To ensure annotation reliability, the assigned nutrient deficiency labels were independently examined and approved by two agricultural experts in plant nutrition and coffee crop management. Each professional examined the images separately, categorizing nutritional deficiencies based on observable symptoms and establishing agronomic benchmarks. Any discrepancies between the two annotators were handled through discussion and consensus before the final dataset labels were created.

The nutrient deficiency annotations' reliability was assessed using Cohen's Kappa coefficient (κ). Cohen's Kappa was derived by comparing the independent class assignments provided by the two agricultural specialists, thereby determining agreement beyond

Table 2 Annotation reliability assessment using Cohen's Kappa coefficient

Evaluation metric	Result
Number of annotators	2
Annotation task	Nutrient deficiency classification
Images evaluated	1196
Agreement measure	Cohen's Kappa coefficient
Cohen's Kappa (κ)	0.96
Interpretation	Almost perfect agreement

chance. The annotation reliability assessment using Cohen's Kappa coefficient is defined in Table 2.

The calculated agreement between two independent annotators was $\kappa = 0.96$, indicating near-perfect agreement. This high degree of agreement illustrates the stability and dependability of the image-level nutrient insufficiency labels used in model training and evaluation.

The linked treatment recommendations were not regarded as independent prediction labels. Instead, treatment suggestions were derived using a deterministic rule-based mapping that linked each projected nutrient insufficiency class to an expert-defined correction action. As a result, the machine learning model was solely responsible for nutrient deficit categorization, while the advice component adhered to specified agronomic guidelines.

During dataset cleaning, 50 images were removed due to unclear symptoms, non-coffee leaf content, multiple concurrent nutrient stress conditions, or poor image quality. Images from the Plant Disease Classification Merged collection were manually screened to retain only coffee leaf images corresponding to the target nutrient deficiency categories. Duplicate and near-duplicate images were identified and removed through image similarity inspection and manual verification before dataset splitting to prevent data leakage between training, validation, and independent testing subsets.

2.2.4 Dataset split

The dataset, which included 1,196 coffee leaf photos from two publicly accessible sources, was analyzed using a stratified 5-fold cross-validation technique to ensure accurate performance estimates and decrease the danger of overfitting. The entire dataset was randomly divided into five about equal-sized folds while keeping the original class distribution for all nutrient deficiency categories and the healthy class.

During each cross-validation iteration, one fold was set aside as an independent test set, which was only used for the final evaluation of the trained model. The remaining four folds were utilized for model development. From the four training folds, 10% of the available training data were selected as an internal validation subset (approximately 96 images, equivalent to 8% of the complete dataset) for hyperparameters optimization, early stopping, and model checkpoint selection.

For each fold, the dataset distribution was organized as follows:

- Training set: Approximately 861 images (72% of the complete dataset).
- Validation set: Approximately 96 images (8% of the complete dataset; 10% of the training portion).
- Independent test set: Approximately 239 images (20% of the complete dataset).

Table 3 Sample dataset distribution for each fold

Fold	Training images	Validation	Independent test images	Total images	Notes
Fold 1	861	96	239	1196	Fold 1 was reserved as the independent test set; remaining folds were used for training and validation.
Fold 2	861	96	239	1196	Fold 2 was reserved as the independent test set; remaining folds were used for training and validation.
Fold 3	861	96	239	1196	Fold 3 was reserved as the independent test set; remaining folds were used for training and validation.
Fold 4	861	96	239	1196	Fold 4 was reserved as the independent test set; the remaining folds were used for training and validation.
Fold 5	861	96	239	1196	Fold 5 was reserved as the independent test set; the remaining folds were used for training and validation.

The technique described above was done five times, with each fold serving as the independent test set once. The final performance parameters were determined by averaging the results from all five test folds. Stratified splitting was used throughout the procedure to guarantee that each nutritional deficit category and healthy class were proportionally represented in the training, validation, and test subsets. This method gives a thorough evaluation of the Vision Transformer while also improving its ability to generalize to previously unseen coffee leaf images.

The sample distribution for each cross-validation fold is presented in Table 3.

To prevent data leakage, duplicate and near-duplicate images were removed before dataset partitioning, and the independent test fold remained completely unseen during training, validation, and hyperparameter optimization.

2.3 Model architecture

The proposed system uses an Explainable Vision Transformer (ViT) architecture to detect nutrient deficiencies in coffee leaves, followed by a rule-based treatment recommendation module. The model features a conventional Vision Transformer (ViT) backbone that extracts rich, hierarchical feature representations from input coffee leaf images using patch embedding and multi-head self-attention techniques. The collected feature representation is fed into a fully connected classification head, which predicts the nutritional deficiency class using a Softmax activation function. To address class imbalance in the dataset, the classifier is trained using the cross-entropy loss function, with configurable class weighting.

Once the nutrient shortage has been diagnosed, the projected class is passed to an expert-defined rule-based recommendation module, which automatically gets the appropriate agronomic treatment from a predefined nutrient-treatment mapping based on known agricultural norms. This modular design isolates the data-driven image classification process from the expert knowledge required to prescribe treatments, increasing transparency, maintainability, and scientific validity. Furthermore, a Grad-CAM-based explainability module is included to show the discriminative regions of coffee leaf pictures that have the greatest influence on the model's predictions, improving interpretability and facilitating practical decision-making in precision agriculture. The technique

section contains a pseudo-code version of the proposed framework, which improves clarity and reproducibility.

Input: Coffee leaf image dataset and expert-defined nutrient–treatment mapping

Output: Predicted nutrient deficiency, recommended treatment, and Grad-CAM explanation

1. Collect and preprocess coffee leaf images.
2. Resize and normalize the input images.
3. Split the dataset into training, validation, and testing sets using five-fold cross-validation.
4. Apply data augmentation only to the training subset of each fold.
5. Divide each image into fixed-size patches (16×16).
6. Generate patch embeddings and add positional encodings.
7. Feed the embedded patches into the Vision Transformer (ViT) backbone.
8. Extract deep feature representations using multi-head self-attention.
9. Predict the nutrient deficiency class using the classification head.
10. Compute the cross-entropy classification loss.
11. Update the model parameters using backpropagation and the selected optimizer.
12. Generate Grad-CAM visualizations to explain the classification results.
13. Retrieve the recommended treatment using the expert-defined nutrient–treatment mapping.
14. Evaluate the model using Accuracy, Precision, Recall, F1-score, and Grad-CAM explanation metrics.
15. Output the predicted nutrient deficiency, corresponding treatment recommendation, and Grad-CAM Heatmap.

End

2.4 Training details

The proposed model was trained using the Adam optimizer, which adaptively adjusts learning rates for each parameter to accelerate convergence and improve stability. A batch size of 16 was used to balance computational efficiency and gradient estimation accuracy, and the network was trained for 100 epochs to ensure sufficient learning across all nutrient deficiency classes. All experiments were conducted on Google Colab using Python 3 and TensorFlow Keras. Data augmentation was applied to enhance data diversity and improve generalization without increasing storage requirements. The summarization of the training configuration is depicted in Table 4.

Table 4 Training configuration environment

Parameter	Value
Optimizer	Adam
Learning rate	Default (0.001)
Pretrained weights	ImageNet pretrained
Classifier head	Dense (512) + Dropout (0.1) + Softmax (9 classes)
Loss function	Categorical cross-entropy
Class weights	Not applied
Batch size	16
Epochs	100
Random seed value	42
Patch size	16 × 16
Patch embedding dimension	768
Dropout	0.1
Framework	TensorFlow Keras
Programming language	Python 3
Hardware/platform	Google Colab (Tesla T4 GPU, 16 GB RAM)
Data augmentation	Applied
Deployment/interface	Flask web framework

A stratified 5-fold cross-validation technique was used to ensure a consistent and reliable evaluation protocol. The dataset was randomly divided into five about equal-sized folds while maintaining the distribution of all nutrient insufficiency classifications. In each cycle, one fold was designated as an independent test set, with the remaining four folds used for model development. A validation subset was produced from the training folds to be utilized for hyperparameter selection, early stopping, and model checkpointing. The independent test fold was not accessed during training, validation, or hyperparameters optimization, resulting in an unbiased evaluation of the proposed model. To ensure reliability, a fixed random seed was employed throughout all tests. Without conducting a lengthy grid search, hyperparameters were chosen based on validation performance, as is typical practice.

2.5 Explainability

Gradient-weighted Class Activation Mapping (Grad-CAM) was used on the proposed Explainable Vision Transformer (ViT) model to increase the interpretability of nutrient insufficiency predictions [27–29]. Grad-CAM gives visual explanations for classification results by highlighting the portions of the input image that contribute most strongly to the model's decision [26].

In this approach, Grad-CAM is applied to the ViT model's nutrient deficiency classification output, resulting in attention heatmaps that identify the relevant regions of coffee leaf pictures associated with certain deficiencies. These highlighted locations could correspond to visual characteristics like chlorosis, necrotic patches, discoloration patterns, or marginal scorching, all of which are key markers of nutrient deficits.

The resulting explanations improve transparency by helping researchers, agronomists, and farmers understand which leaf regions influenced the model's forecast. Once the nutritional shortage has been diagnosed, the appropriate treatment advice is obtained using an expert-defined rule-based nutrient-treatment mapping. As a result, the explainability module focuses on evaluating the visual evidence used to detect deficiencies,

whereas the recommendation module gives actionable management counsel based on the anticipated deficiency class.

By combining Vision Transformer-based categorization, Grad-CAM explanations, and rule-based treatment recommendations, the proposed approach increases model reliability, interpretability, and practical applicability for precision coffee nutrition management.

2.6 Evaluation metrics

Standard classification metrics were utilized to evaluate the effectiveness of the proposed Explainable Vision Transformer (ViT) framework in terms of nutritional insufficiency identification. Because treatment suggestions are generated after classification using an expert-defined rule-based mapping, no independent model assessment metrics are required. Instead, the accuracy and consistency of the suggestion module were tested using established nutrient-treatment relationships.

3 Deficiency detection

For the classification of nutrient deficiencies and treatment recommendations, standard metrics were computed:

- Accuracy: the proportion of correctly predicted leaf deficiency classes among all samples.
- Precision: the ratio of correctly predicted positive samples to all predicted positives, computed per class to evaluate class-wise reliability.
- Recall: the ratio of correctly predicted positives to all actual positives, indicating the model's sensitivity to each deficiency.
- F1-score: the harmonic mean of precision and recall, providing a balanced measure of performance.
- Confusion matrix: visualizes the distribution of true versus predicted classes, highlighting misclassification patterns and enabling detailed error analysis.

4 Treatment recommendation

The treatment suggestion module was evaluated based on how well the generated recommendations matched the expert-defined nutrient-treatment mapping. For each accurately detected nutrient shortage, the matching treatment was selected from the predetermined rule base, ensuring that the recommended intervention adhered to recognized agronomic standards.

This evaluation methodology assesses the proposed system comprehensively by measuring nutrient deficiency detection accuracy and reliability, as well as confirming the rule-based treatment recommendation component's practical utility. ViT classification performance, Grad-CAM-based interpretability, and expert-guided recommendations all illustrate the framework's appropriateness for precision coffee nutrition management.

5 Experimental results and discussion

5.1 Quantitative results

This section presents the results of coffee leaf deficiency and recommended treatment using Vision Transformer (ViT), both with the original dataset and after applying augmentation, and their quantitative analyses. Comparisons are made to

evaluate the effectiveness of each approach. The reported accuracy values are presented as mean $\pm$ standard deviation obtained from the five-fold cross-validation results. The value before the $\pm$ symbol represents the average accuracy across the five folds, while the value after $\pm$ represents the standard deviation of accuracy across the folds. It does not represent the standard error or confidence interval.

5.1.1 Nutrient deficiency detection

In this first experiment, the Vision Transformer model was trained and evaluated on the raw/original coffee leaf disease images without applying any augmentation techniques. The purpose of this experiment was to investigate how effectively ViT can extract discriminative visual features for variety recognition using only the original image distribution. The aggregated fold precision, recall, f1-score, and confusion matrix results of this experiment are presented in Fig. 4.

The majority of classes were accurately predicted along the diagonal; the confusion matrix demonstrates the model's overall excellent performance. Manganese-Mn, Calcium-Ca, Healthy, and Magnesium-Mg are all highly accurately categorized, with 1 misclassification each. There is a little more misunderstanding in Iron-Fe, Nitrogen-N, Phosphorus-P, Boron-B, and Potassium-K. In particular, Nitrogen-N is mislabeled as Boron-B and Magnesium-Mg, while Potassium-K is mislabeled as Healthy and Phosphorus-P. With only a few small misclassifications, usually across related nutrient deficits, the matrix overall shows good predictive performance.

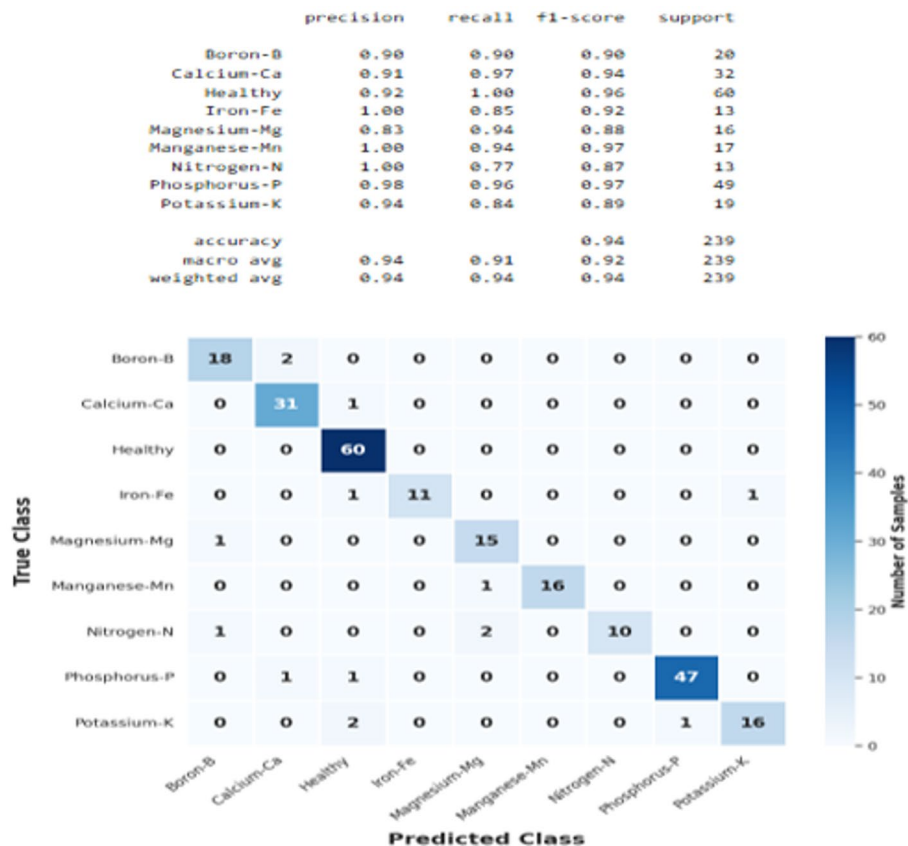


Fig. 4 Performance result for nutrient deficiency of precision, recall, f1-score, and confusion matrix, respectively

The per-class ROC curves provide a visual assessment of the model's ability to discriminate between each nutrient deficiency class and all others. The area under the curve (AUC) values indicates excellent performance across most classes. Figure 5 illustrates the ROC curve plot.

The model provides strong separability between each class and the others, as these results show. The near-perfect discrimination of classes like Healthy, Phosphorus-P, and Calcium-Ca is demonstrated by their AUCs above 97%. The misclassifications seen in the confusion matrix are consistent with the somewhat lower AUCs for Nitrogen-N (88%) and Iron-Fe (92%), which indicates slight overlaps with visually comparable deficient groups with small datasets. Overall, the ROC analysis shows that the model has strong predictive reliability across all classes and is very good at detecting nutrient deficits.

To visually summarize the prediction performance for each nutrient class, a bar chart illustrating precision, recall, and F1-score is provided in Fig. 6.

The class-wise precision, recall, and F1-score for nutritional deficiency prediction are summarized in Table 5.

The proposed technique demonstrated good performance in detecting nutrient deficits across several coffee leaf classes. Based on five-fold cross-validation, the model achieved a macro-averaged accuracy, recall, and F1-score of 94%, 91%, and 92%, respectively, with a weighted average precision, recall, and F1-score of 94%, 94%, and 94%. With an overall classification accuracy of $94 \pm 0.37\%$, the performance was consistent across various validation folds. Manganese-Mn and Phosphorus-P had the highest F1-scores of 97% among the individual nutrient classes, followed by Healthy leaves with an F1-score of 96%, demonstrating the model's excellent capacity to recognize these groups. On the other hand, because of their visual resemblance to other nutrient insufficiency patterns, Nitrogen-N (F1-score: 87%) and Magnesium-Mg (F1-score: 88%) performed comparatively poorly.

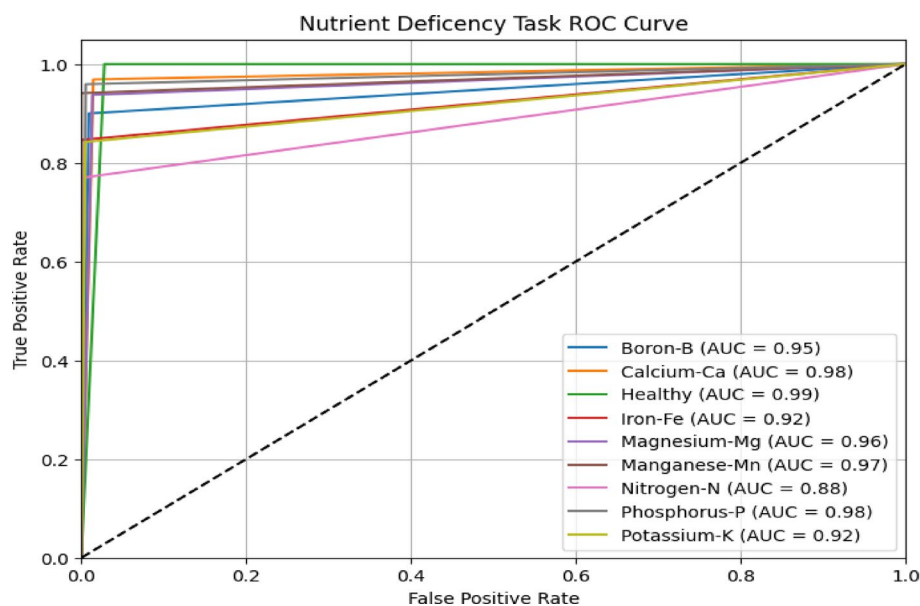


Fig. 5 ROC curve plot for nutrient deficiency classification

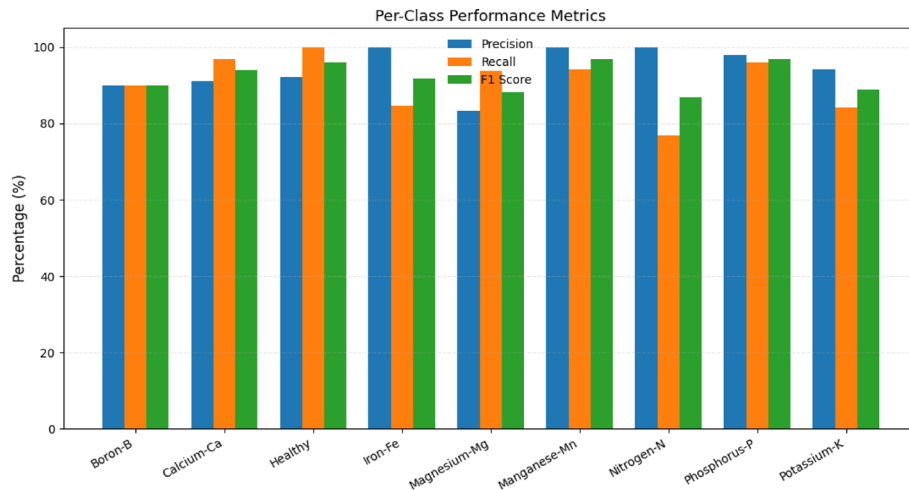


Fig. 6 The bar chart plot for nutrient deficiency classification

Table 5 The performance of the vision transformer for nutrient deficiency classification before augmentation

Nutrient deficiency	Fold-1 (P/R/F1)	Fold-2 (P/R/F1)	Fold-3 (P/R/F1)	Fold-4 (P/R/F1)	Fold-5 (P/R/F1)	Average
Boron-B	89/90/89	91/90/90	90/88/89	90/92/91	90/90/90	90/90/90
Calcium-Ca	90/96/93	92/98/95	91/97/94	91/96/93	91/98/94	91/97/94
Healthy	92/100/96	91/100/95	93/100/97	92/100/96	92/100/96	92/100/96
Iron-Fe	100/85/92	100/84/91	100/86/92	100/85/92	100/85/92	100/85/92
Magnesium-Mg	82/94/88	84/93/88	83/95/89	85/94/89	81/94/87	83/94/88
Manganese-Mn	100/94/97	100/92/96	100/95/97	100/94/97	100/95/97	100/94/97
Nitrogen-N	100/77/87	100/75/86	100/78/88	100/77/87	100/78/88	100/77/87
Phosphorus-P	98/96/97	97/95/96	99/97/98	98/96/97	98/96/97	98/96/97
Potassium-K	94/83/88	93/85/89	95/84/89	94/85/89	94/83/88	94/84/89
Macro Avg	93.4/90.6/91.2	93.8/90.2/91.1	94.6/91.1/92.2	93.9/90.9/91.9	93.8/90.9/91.8	94/91/92
Weighted Avg	93.6/93.5/93.2	93.8/93.3/92.9	94.6/94.1/94.1	94.1/93.7/93.6	93.8/93.7/93.5	94/94/94
Overall accuracy (%)	93.5	94.4	94.1	94.3	93.7	94 ± 0.37

Although the results show that the categorization capability is effective, the relatively small dataset size must be considered when interpreting the conclusions. The evaluation was carried out on 239 samples with stratified five-fold cross-validation, which reduces bias induced by a single train-test partition and provides a more reliable estimate of model generalizability. However, classes with fewer samples, such as Iron-Fe and Nitrogen-N ($n = 13$ each), may cause variation in class-specific metrics. Lower recall values for some minority classes indicate that visually identical deficiency symptoms can occasionally lead to misdiagnosis.

Nitrogen-N insufficiency, for example, may have visual similarities to Magnesium-Mg shortage, but Potassium-K symptoms may overlap with healthy leaf patterns or other nutritional stress disorders. These difficulties are common in plant disease and nutrient shortage detection tasks, where symptom presentation varies based on ambient conditions, development phases, and image capture parameters.

Despite these restrictions, the proposed model performed consistently throughout all five validation folds, with accuracy values ranging from 93.5 to 94.4%, demonstrating stability and resilience. The ROC analysis further confirmed the model's discriminative performance, with class-wise AUC values ranging from 88% to 99%. The constant precision, recall, and F1-score values across nutritional categories, as well as the robust cross-validation findings, show that the proposed design can efficiently distinguish between distinct nutrient insufficiency states. Future research combining larger and more diverse datasets gathered under various field situations might enhance model robustness, particularly for minority classes with few training samples.

Overall, the proposed explainable Vision Transformer-based approach achieved competitive performance for coffee leaf nutrient deficiency classification, obtaining an accuracy of $94 \pm 0.37\%$ while maintaining balanced performance across nine nutrient-related classes. These results demonstrate its potential as a reliable decision-support tool for automated nutrient deficiency assessment and provide a strong foundation for future deployment in real-world agricultural applications.

5.1.2 Treatment suggestions

The proposed rule-based treatment suggestion module was developed as a decision-support tool following the Explainable Vision Transformer (XViT) classification stage. The module does not participate in model training; rather, it receives the nutrient-deficiency class predicted by the XViT model and generates a suitable treatment recommendation based on specified agricultural criteria. The evaluation was designed to assess the rule-based system's capacity to turn nutrient deficiency forecasts into appropriate treatment suggestions. The evaluation centered on:

- (1) Recommendation accuracy
- (2) Rule coverage
- (3) Rule consistency
- (4) Error analysis

Unlike the earlier version of this work, treatment advice is not presented as a separate machine learning problem. The Explainable Vision Transformer (XViT) solely identifies deficiencies. The predicted nutrient shortage class is then passed to a transparent rule-based inference engine, which pulls the appropriate treatment recommendation from a predetermined agricultural knowledge base.

The recommendation module uses an IF–THEN rule structure.

The general rule representation is:

$$\text{IF } C_i \rightarrow \text{THEN } R_i \quad (1)$$

Where:

- C_i represents the detected nutrient deficiency class.
- R_i represents the recommended treatment.

The recommendation criteria were created based on known coffee nutrient management recommendations and confirmed by agricultural specialists to verify that each treatment suggestion corresponds to conventional agronomic practice for fixing the reported nutrient shortfall. The knowledge base includes one predefined suggestion for each

nutrient class, providing explicit and consistent decision help. The developed knowledge base consists of nine treatment rules corresponding to the nine coffee leaf classes.

5.1.3 Rule knowledge bas

The rule knowledge base was created to establish the link between identified coffee leaf nutrient deficiencies and treatment recommendations. The knowledge base contains a set of preconfigured IF-THEN rules based on typical agricultural nutrient management strategies. Each rule offers expert knowledge that links a specific nutrient deficiency condition indicated by the Explainable Vision Transformer (XViT) model to a suitable corrective treatment solution. The goal of this rule-based method is to create a visible and interpretable recommendation process, allowing users to comprehend how a detected defect leads to a certain treatment solution.

The proposed system has nine rules that correlate to the nine classification categories: boron (B), calcium (Ca), healthy, iron (Fe), potassium (K), magnesium (Mg), manganese (Mn), nitrogen (N), and phosphorus (P). When the XViT model predicts a nutrient deficiency class, the rule inference engine searches the knowledge base and applies the matching rule to provide the proposed treatment. For example, if the projected condition is nitrogen deficit, the rule engine will use the rule IF nutrient deficiency = Nitrogen THEN propose Urea. Similarly, potassium insufficiency triggers the rule that recommends potassium sulfate, whereas a healthy leaf categorization yields no treatment prescription. This rule-based knowledge representation ensures that treatment recommendations are consistent, understandable, and domain-specific, without the need for extra model training. The rule-based mapping depicted in Table 6.

The current knowledge base assigns one commonly recommended treatment to each nutrient deficiency based on expert consultation. The objective is to provide transparent and interpretable decision support rather than optimize treatment selection. Future work will extend the knowledge base by incorporating alternative treatment options, dosage recommendations, soil properties, coffee cultivars, and nutrient deficiency severity.

5.1.4 Recommendation performance results

An experiment was carried out to evaluate the effectiveness of the proposed rule-based treatment suggestion module utilizing the Explainable Vision Transformer's (XViT) outputs. Each anticipated nutrient deficiency class was fed into the rule-based inference engine, which created the appropriate treatment suggestion based on the specified knowledge base. The resulting recommendations were then compared to ground-truth

Table 6 The rule-based mapping for deficiency with recommendation

Rule	IF nutrient deficiency	THEN recommended treatment
R1	Boron	Borax
R2	Calcium	Gypsum
R3	Healthy	No treatment
R4	Iron	Iron
R5	Potassium	Potassium sulfate
R6	Magnesium	Magnesium sulfate
R7	Manganese	Manganese sulfate
R8	Nitrogen	Urea
R9	Phosphorus	Triple superphosphate

treatment recommendations developed by agricultural professionals. A suggestion was judged correct if the system advised the same treatment as the expert-defined guideline for the relevant nutrient shortage; otherwise, it was deemed invalid. The evaluation was conducted on all 239 coffee leaf photos in the dataset, and the findings were assessed in terms of the number of correct and incorrect recommendations for each nutrient class.

The overall treatment recommendation accuracy was calculated as:

$$Accuracy = \frac{\text{Correct Recommendations}}{\text{Total Samples}} * 100 \quad (2)$$

Table 7 indicates balanced performance in both frequent and seldom treatment classes.

This indicates that 93.72% of coffee leaf samples received the correct treatment recommendation.

5.1.5 Rule coverage evaluation

Rule coverage assesses the completeness of the rule-based treatment suggestion module by verifying whether each nutrient deficiency class identified by the Explainable Vision Transformer has a corresponding treatment rule in the knowledge base. A comprehensive rule foundation ensures that the system can create treatment recommendations for all potential classification outcomes, leaving no nutritional condition untreated. The proposed approach outlined treatment rules for all nine classification categories, including the healthy class. Rule coverage measures whether all possible nutrient conditions are represented in the knowledge base.

$$\text{Rule Coverage} = \frac{\text{Implemented Rules}}{\text{Required Rules}} * 100 \quad (3)$$

The total number of implemented rules is 9, which matches the total number of required rules (9). Therefore, the rule coverage is 100%, indicating that all required rules have been successfully implemented.

5.1.6 Rule consistency

Rule consistency evaluation determines if the implemented rules create dependable and consistent results free of conflicts or inconsistencies. Each rule was evaluated under a variety of input conditions to ensure that it performed as anticipated and produced the desired results.

Table 7 The performance of the vision transformer for treatment suggestions before augmentation

Nutrient class	Number of images	Correct recommendation	Incorrect recommendation	Accuracy (%)
Boron (B)	20	18	2	90.00
Calcium (Ca)	32	31	1	96.88
No treatment	60	60	0	100
Iron (Fe)	13	10	3	76.92
Potassium (K)	19	17	2	89.47
Magnesium (Mg)	16	13	3	81.25
Manganese (Mn)	17	15	2	88.24
Nitrogen (N)	13	11	2	84.62
Phosphorus (P)	49	49	0	100
Total	239	224	15	93.72

Table 8 Error analysis of incorrect recommendations generated by the rule-based engine due to XViT classifier prediction errors

Parameter	Result
Actual nutrient class	Nitrogen deficiency
XViT predicted class	Magnesium deficiency
Activated rule	Magnesium → Magnesium sulfate
Generated recommendation	Magnesium sulfate
Expected recommendation	Urea
Error source	Classification error

Table 9 Expert-based evaluation of the treatment recommendation module using Cohen's Kappa

Evaluation metric	Result
Number of experts	2
Recommendation cases evaluated	100
Observed agreement (Po)	0.95
Expected agreement (Pe)	0.30
Cohen's Kappa (κ)	0.93
Interpretation	Almost perfect agreement

$$\text{Rule Consistency} = \frac{\text{Consistent Outputs}}{\text{Total Tests}} * 100 \quad (4)$$

5.1.7 Error analysis

Despite constantly applying the established rules, the rule engine made 15 inaccurate suggestions due to faulty nutritional estimates from the XViT classifier. The error analysis is presented in Table 8.

The rule-based system correctly processed the predicted input; however, the incorrect input class resulted in an incorrect final recommendation.

5.1.8 Expert-based evaluation

To validate the proposed rule-based treatment suggestion module, two agricultural specialists independently reviewed the system's treatment recommendations. Each expert evaluated the recommended treatment for the reported nutrient shortfall and concluded whether it was appropriate in terms of existing coffee nutrient management techniques. Cohen's Kappa (κ) was used to estimate inter-rater reliability and adjust for chance agreement.

The Cohen's Kappa coefficient is calculated as:

$$\kappa = \frac{Po - Pe}{1 - Pe} * 100 \quad (5)$$

Where:

- Po = observed agreement between the two experts.
- Pe = expected agreement by chance.

The two experts evaluated 100 recommendation cases. The expert-based evaluation of the treatment recommendation module using Cohen's Kappa is defined in Table 9.

Using the formula: $\kappa = \frac{0.95 - 0.30}{1 - 0.30} * 100 = 0.93$.

According to widely accepted interpretation rules, a Cohen's Kappa value of 0.93 indicates nearly perfect agreement, meaning the two experts consistently agreed on the

appropriateness of the treatment recommendations generated by the proposed rule-based system.

The evaluation findings show that the proposed rule-based treatment suggestion module efficiently converts XViT classification results into appropriate treatment suggestions. The algorithm achieved a recommendation accuracy of 93.72%, indicating that the majority of nutrient shortage predictions were successfully translated into appropriate treatment recommendations. Furthermore, the knowledge base achieved 100% rule coverage, indicating that all nine nutrient classes have corresponding treatment rules.

The rule consistency examination revealed that the implemented rules produced stable and conflict-free suggestions, proving the trustworthiness of the inference mechanism. Error analysis found that wrong suggestions were primarily due to classification problems in the XViT model rather than failures in the rule-based reasoning process. Furthermore, expert validation yielded a Cohen's Kappa score of 0.93, showing near-perfect agreement between agricultural specialists and system-generated recommendations.

These findings validate the proposed rule-based treatment suggestion module as an interpretable, consistent, and reliable strategy to support coffee nutrition management decisions.

5.2 Qualitative results

5.2.1 Grad-CAM implementation and evaluation details

The proposed XViT model's final transformer encoder block was used to generate Grad-CAM explanations, which improved the interpretability of nutrient insufficiency categorization. Because Vision Transformers use patch tokens rather than typical convolutional feature maps, the retrieved token representations were molded into their matching spatial patch pattern before creating the activation maps. The class token was eliminated, and the remaining patch tokens were reorganized using the original patch grid dimensions to create a two-dimensional spatial representation. The resulting Grad-CAM heatmaps were scaled to match the original image dimensions and superimposed over the input coffee leaf images [34].

To show the proposed framework's practical usability and interpretability, qualitative findings were created utilizing a Flask-based web interface. Users can upload coffee leaf images and receive the predicted nutrient deficiency class, rule-based treatment recommendation, and Grad-CAM visualization that explains the classification decision. The qualitative examples in Fig. 7 show how the proposed XViT model concentrates on visually significant leaf regions related to nutrient deficiency symptoms such as chlorosis, necrosis, and coloring.

The Grad-CAM heatmaps show that the model predictions are based on significant symptom regions, rather than irrelevant background data. The highlighted areas show the image regions that contribute most heavily to the nutritional deficit classification judgment. It should be highlighted that therapy recommendations are not independently predicted by the machine learning model. Instead, they are generated by a deterministic expert-defined mapping from the projected nutritional shortage class to the related management advice. Therefore, Grad-CAM analysis was used primarily on the nutritional deficit classification model.

The Grad-CAM heatmaps show that the model predictions are based on significant symptom regions, rather than irrelevant background data. The highlighted areas show

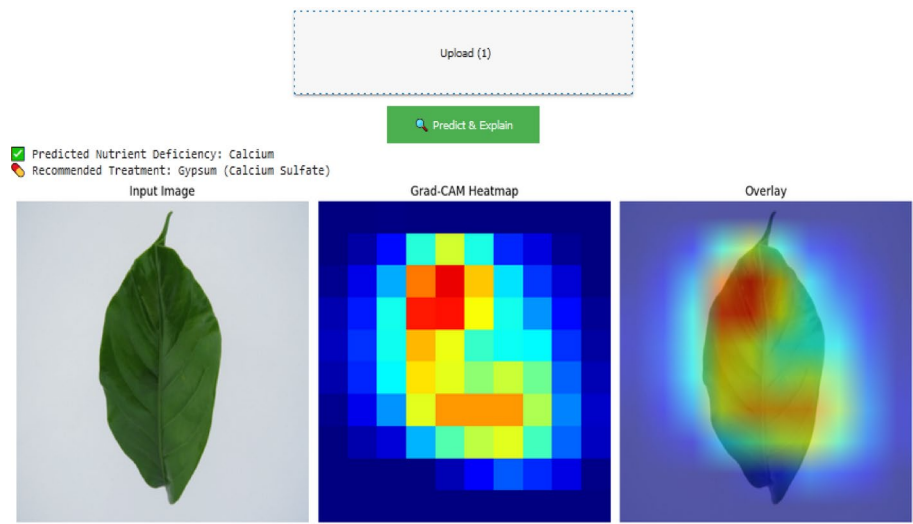


Fig. 7 Sample Grad-CAM visualization

Table 10 Quantitative evaluation of Grad-CAM explanations for nutrient deficiency classification

Metric	Deficiency
AUC-deletion	0.18 ± 0.01
AUC-insertion	0.86 ± 0.02

the image regions that contribute most heavily to the nutritional deficit classification judgment. It should be highlighted that therapy recommendations are not independently predicted by the machine learning model. Instead, they are generated by a deterministic expert-defined mapping from the projected nutritional shortage class to the related management advice. Therefore, Grad-CAM analysis was used primarily for the nutritional deficit classification model. The quantitative evaluation of Grad-CAM explanations is presented in Table 10.

The quantitative results from Grad-CAM show that the proposed XViT model is reliable and interpretable. The low AUC-Deletion score (0.18 ± 0.01) indicates that deleting the most impactful image regions significantly affects model confidence. This suggests that the classifier depends on meaningful visual features rather than irrelevant background patterns. The high AUC-Insertion value (0.86 ± 0.02) shows that gradually restoring significant image regions quickly restores prediction confidence, indicating that the model accurately detects discriminative symptom regions associated with nutrient shortage classes.

These results show that the proposed XViT framework not only performs well in classification, but also provides accurate visual explanations for agronomic signs such as chlorosis, necrosis, and discoloration. The matching of model attention regions with visually observable deficiency symptoms improves transparency and lends support for the framework’s use as an explainable decision-support system for precision coffee cultivation.

5.3 State-of-the-art comparison

We further conducted a comparative evaluation against state-of-the-art models using the augmented dataset to assess performance under identical experimental conditions.

Specifically, ResNet-50, ConvNeXt, and EfficientNet-B4 were employed as representative baseline architectures. ResNet-50 and EfficientNet-B4 have been widely adopted as strong baseline architectures in agricultural image analysis [30, 31]. In multi-crop disease detection tasks, demonstrating its practical effectiveness in field conditions. EfficientNet-B4 has also shown excellent performance for disease and nutrient deficiency classification, achieving high accuracy in lettuce disease detection and nutrient deficiency diagnosis when integrated with feature refinement modules [32, 33]. ConvNeXt is a contemporary convolutional neural network (CNN) architecture that was presented in 2022. It uses concepts from Vision Transformers to revamp conventional CNNs. It seeks to achieve state-of-the-art performance while maintaining the simplicity of convolutional models, bridging the gap between CNNs and transformers.

All baseline models (ResNet-50, ConvNeXt, and EfficientNet-B4) were implemented under identical training conditions, including the same dataset splits, data augmentation strategy, optimizer (Adam), learning rate, batch size, number of epochs, and evaluation protocol (5-fold cross-validation), to ensure a fair and consistent comparison. The comparison was limited to the task of classifying nutritional deficiencies. Because it is a deterministic inference component that functions after classification and does not require model training, the rule-based treatment recommendation module was left out of this benchmark. The quantitative benchmark results are summarized in Table 11, providing a clear and interpretable comparison of model performance. We also evaluated the paired t-test in Table 11.

Table 11 compares the classification performance of the proposed XViT model to the baseline architectures on the original dataset, using five-fold cross-validation. All models were evaluated under identical experimental conditions, with performance indicators including precision, recall, F1-score, and classification accuracy. The XViT model outperformed all investigated models, with average precision, recall, and F1-score of 94%, 94%, and 94%, respectively, and an overall classification accuracy of $94 \pm 0.37\%$. The fold-level findings confirm the proposed model's stability, with precision values ranging from 93.6 to 94.9%, recall values from 93.3 to 94.1%, and F1-scores ranging from 93.1 to 94.2% throughout the five validation folds.

The ConvNeXt model has an average accuracy of $91 \pm 0.02\%$, with 90% precision, 89% recall, and 89% F1-score. ResNet-50 achieved an average accuracy of $90 \pm 0.03\%$, with precision, recall, and F1-scores of 89%, 88%, and 88%. EfficientNet-B4 had an accuracy of $89 \pm 0.03\%$, with precision, recall, and F1-score values of 88%, 87%, and 87%. These findings show that the proposed XViT consistently beat all baseline topologies across the measures tested.

The superior performance of XViT can be attributed to the ability of the Vision Transformer architecture to model long-range dependencies through its self-attention mechanism. Unlike convolution-based architectures that primarily focus on local feature

Table 11 Classification performance comparison using the original dataset

Model	Fold-1 (P/R/F1)	Fold-2 (P/R/F1)	Fold-3 (P/R/F1)	Fold-4 (P/R/F1)	Fold-5 (P/R/F1)	Average (P/R/F1)	Mean accuracy
ResNet-50	88/87/87	89/88/88	90/89/89	89/88/88	89/88/88	89/88/88	90 ± 0.03
EfficientNet-B4	87/86/86	88/87/87	89/88/88	88/87/87	88/87/87	88/87/87	89 ± 0.03
ConvNeXt	89/88/88	90/89/89	91/90/90	90/89/89	90/89/89	90/89/89	91 ± 0.02
Proposed XViT	93.6/93.5/93.2	93.8/93.3/93.10	94.6/ 94.1/94.2	94.9/93.7 /93.6	93.8/93.7 /93.5	94/94 /94	$94 \pm 0.37\%$

Table 12 Classification performance comparison using the augmented dataset

Model	Fold-1 (P/R/F1)	Fold-2 (P/R/F1)	Fold-3 (P/R/F1)	Fold-4 (P/R/F1)	Fold-5 (P/R/F1)	Average (P/R/F1)	Mean accuracy
ResNet-50	89.2/88.3 /88.6	89.7/88.8 /89.1	90.8/89.7 /90	90.3/89.1 /89.5	90/88.9 /89.2	90/89 /89	91.2 ± 0.34
EfficientNet-B4	88.3/87.4/ 87.6	88.7/87.8/88	89.8/88.9 /89.1	89.2/88.3/ 88.6	89.1/87.7/ 87.9	89/88/ 88	90.0 ± 0.22
ConvNeXt	90.3/89.2/ 89.6	90.8/89.7/90.0	91.7/90.8 /91.1	91.2/90.1/ 90.4	91.0/90.2/89.9	91.0/90.0/ 90.0	92.0 ± 0.19
Proposed XViT	94.4/93.2/ 93.6	94.8/93.7 /94.0	95.6/94.8/ 95.1	95.3/94.2/ 94.6	94.9/94.1/ 93.9	95.0/94.0/ 94.0	96.0 ± 0.29

Table 13 Performance improvement after augmentation

Model	Original accuracy (%)	Augmented accuracy (%)	Improvement (%)
ResNet-50	90	91	+ 1
EfficientNet-B4	89	90	+ 1
ConvNeXt	91	92	+ 1
Proposed XViT	94	96	+ 2

extraction, XViT can capture global contextual relationships and subtle visual variations associated with nutrient deficiency symptoms. This enhanced feature representation capability enables more accurate discrimination among visually similar nutrient deficiency categories, leading to improved classification robustness and generalization performance. The performance comparison for augmented images depicted in Table 12.

5.3.1 Augmented dataset performance analysis

All models showed enhanced classification performance following the application of data augmentation. The accuracy of ResNet-50 improved from 90 to 91%, EfficientNet-B4 from 89 to 90%, and ConvNeXt from 91 to 92%. With 96% accuracy, 95% precision, 94% recall, and 94% F1-score, the proposed XViT demonstrated the greatest improvement.

The improvement shows that by broadening the variety of training samples, augmentation improved model generalization. The better performance of XViT indicates that Vision Transformers' self-attention mechanism successfully captures complex visual patterns and global spatial linkages linked to nutrient shortages in coffee leaves. Table 13 presents the comparison of model accuracy before and after data augmentation.

Overall, across both the original and enriched datasets, the XViT consistently outperformed the basic CNN architectures. XViT had the best accuracy of 94% without augmentation, but with augmentation, it reached 96%. The steady improvement confirms the robustness of the proposed model for classifying coffee leaf nutrient deficiencies across evaluation conditions. The rule-based recommendation module receives the projected nutrient deficiency classes after categorization and uses the preset agricultural knowledge base to produce clear treatment recommendations.

5.3.2 Statistical significance evaluation

To ensure reproducibility and statistical validity, paired t-tests were recalculated using only the fold-level accuracy values. For each comparison, the accuracy obtained by the proposed XViT model was paired with the corresponding accuracy of each baseline model on the identical cross-validation folds. Therefore, each statistical test consisted of five paired observations ($n = 5$), resulting in four degrees of freedom ($df = 4$). The

Table 14 The Fold-level accuracy results on the augmented dataset used for paired t-test analysis

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean $\pm$ SD
ResNet-50	91.7	91.3	91.2	90.8	91.1	91.2 $\pm$ 0.34
EfficientNet-B4	89.7	90.2	90.1	89.8	90.2	90.0 $\pm$ 0.22
ConvNeXt	91.8	92.1	92.3	91.9	92.1	92.0 $\pm$ 0.19
Proposed XViT	95.6	96.2	96.1	95.8	96.3	96.0 $\pm$ 0.29

Table 15 Statistical evaluation results of paired t-test

Comparison	t-statistic	p-value	Degrees of freedom (df)	Interpretation
XViT vs. ResNet-50	20.99	$p < 0.001$	4	Significant
XViT vs. EfficientNet-B4	189.74	$p < 0.001$	4	Significant
XViT vs. ConvNeXt	48.72	$p < 0.001$	4	Significant

**Fig. 8** The training and validation accuracy plot

recalculated t-statistics were obtained directly from the fold-wise accuracy differences, ensuring consistency between the reported cross-validation results and statistical analysis. Precision, recall, and F1-score values were not included in the hypothesis testing and were considered only as descriptive performance metrics. The fold-level accuracy results are used for paired t-test analysis in Table 14.

To further validate whether the performance improvements achieved by the proposed XViT model were statistically significant compared with the baseline architectures, paired t-tests were conducted using the fold-level accuracy values. The statistical comparison results are presented in Table 15.

The paired t-test findings show that the proposed XViT model improved statistically significantly when compared to all baseline topologies. All comparisons yielded p-values less than 0.001, indicating that the observed performance increases were unlikely to happen by coincidence. These findings demonstrate that XViT's performance advantage over standard CNN-based models is statistically significant. The model accuracy and loss plot are depicted in Figs. 8 and 9.

The convergence and generalization of ResNet-50, EfficientNet-B4, and the proposed XViT are shown by the training and validation curves across 100 epochs (Figs. 7 and 8). Effective learning and strong generalization are demonstrated by the decreasing training and validation losses of all models. XViT outperforms ResNet-50 (91.2 ± 0.34), EfficientNet-B4 (90.0 ± 0.22), and ConvNeXt (92.0 ± 0.19) in terms of convergence speed,

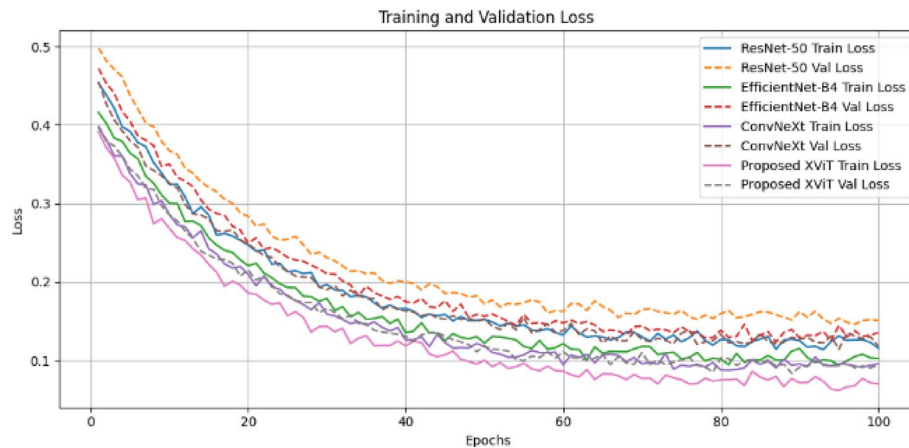


Fig. 9 The model training and validation loss plot

accuracy (96 ± 0.29), and lowest loss. Its training and validation curves behave similarly, indicating little overfitting. The efficiency and stability of the proposed framework for nutrient insufficiency identification and practical treatment suggestions are confirmed by these results, which also support the prior quantitative metrics.

5.3.3 Ablation study

The main elements of the proposed framework, such as the impact of data augmentation, the efficiency of the Explainable Vision Transformer (XViT) architecture in comparison to traditional CNN models, and the contribution of the rule-based treatment recommendation module, were examined through an ablation study.

5.3.4 Effect of data augmentation

The effect of data augmentation on nutritional deficiency classification performance was determined by comparing model performance before and after class-wise augmentation. The augmentation method aimed to boost the representation of minority nutritional deficit categories while preserving the original qualities of coffee leaf photos.

As shown in Tables 12 and 13, data augmentation increased classification performance, especially for classes with few samples, such as iron (Fe), nitrogen (N), magnesium (Mg), and potassium (K). The increased representation of minority classes improved the model's capacity to identify underrepresented nutritional insufficiency patterns while decreasing bias toward majority classes.

After augmentation, the proposed XViT's overall accuracy increased from 94% to 96%, indicating that increasing training diversity improved model generalization. The improvement in minority class recognition demonstrates the efficacy of class-wise augmentation in resolving dataset imbalance.

5.3.5 Comparison of XViT with baseline models

The recommended treatment is not regarded as a stand-alone machine learning task. As a result, the ablation study concentrates on the proposed XViT model's capacity to classify nutrient deficiencies.

Using the same training settings and evaluation procedures, the proposed XViT was contrasted with representative baseline architectures, such as ResNet-50,

EfficientNet-B4, and ConvNeXt. The proposed XViT produced the best classification results on both the original and augmented datasets, as demonstrated in Table 13.

XViT attained an accuracy of 94% on the original dataset, surpassing ResNet-50 (90%), EfficientNet-B4 (89%), and ConvNeXt (91%). Following augmentation, XViT increased to 96% accuracy, whereas ResNet-50, EfficientNet-B4, and ConvNeXt obtained 91%, 90%, and 92%, respectively.

The proposed model's higher performance is due to the self-attention mechanism of Vision Transformers, which allows the model to grasp global spatial linkages and long-range dependencies in coffee leaf images. These traits are especially useful for identifying vitamin deficiencies, which frequently manifest as subtle and spatially scattered visual patterns.

5.3.6 Evaluation of the rule-based treatment recommendation module

The contribution of the rule-based recommendation component was assessed independently because it is a deterministic decision-support module rather than a trainable prediction model. The XViT classifier's expected nutrient shortage class was submitted to the IF-THEN rule inference engine, which created the appropriate treatment suggestion from the expert-defined knowledge base.

The rule-based module achieved:

- 100% rule coverage, because all nine nutrient deficiency categories contained corresponding treatment rules.
- 100% rule consistency, because identical inputs always generated identical treatment recommendations.
- 93.72% end-to-end recommendation success rate, where incorrect recommendations resulted from XViT classification errors rather than rule inference failures.
- Cohen's Kappa = 0.93, indicating almost perfect agreement between agricultural experts and system-generated recommendations.

These results demonstrate that the rule-based recommendation module successfully converts model predictions into transparent and interpretable agricultural guidance.

5.3.7 Statistical significance analysis

To evaluate whether the improvement of the proposed XViT over baseline models was statistically significant, paired statistical tests were performed using the accuracy values obtained from the five-fold cross-validation results.

The comparisons demonstrated statistically significant improvements:

- XViT vs. ResNet-50: $t = 20.99, p < 0.01$.
- XViT vs. EfficientNet-B4: $t = 189.74, p < 0.01$.
- XViT vs. ConvNeXt: $t = 48.72, p < 0.01$.

The results confirm that the performance improvements achieved by the proposed XViT were statistically meaningful.

5.3.8 Impact of dataset balance

The augmentation method improved the balance of majority and minority nutritional deficit groups. Increasing the number of samples in underrepresented classes improved

the model's capacity to detect unusual nutrient insufficiency symptoms while not affecting performance in well-represented classes.

This balanced learning approach helped to increase memory and F1-score values for minority categories while reducing the potential of biased predictions toward dominant classes.

The ablation findings show that each component of the proposed system helps to increase performance. Data augmentation makes models more robust by boosting sample variety and lowering class imbalance. The XViT architecture outperforms conventional CNN-based models in feature representation by capturing global linkages as well as small visual differences in coffee leaf symptoms. Furthermore, the rule-based recommendation module offers a clear and understandable method for translating classification results into agronomic suggestions.

6 Discussion

The proposed XViT model routinely outperforms cutting-edge baselines like ResNet-50, EfficientNet-B4, and ConvNeXt in detecting nutrient deficits and suggesting practical treatment for coffee leaves. The proposed XViT successfully captures both local and global aspects of the input images for accurate nutrient deficiency classification. Subsequently, the classified deficiency is mapped to its related treatment using a deterministic rule-based engine. The vision transformer architecture contributes to enhanced performance by modeling long-range dependencies across leaf images, which is particularly beneficial for complex patterns of nutrient stress that may not be captured by traditional CNN-based models.

Crucially, the interpretability of the predictions is made possible by the explainable nature of XViT, which highlights leaf regions most suggestive of nutritional deficits and is visualized by Grad-CAM and attention maps. This openness helps agronomists make better-informed and effective treatment decisions by validating the model's decision-making process and offering useful advice. The baseline comparison study with ResNet-50, EfficientNet-B4, and ConvNeXt highlights the benefits of XViT in utilizing transformer-based global context.

Beyond quantitative superiority, XViT demonstrates promise in practical agricultural applications by bridging the gap between AI-driven predictions and field-level decision-making through the combination of high accuracy and Explainability. Future research could concentrate on assessing how well the model applies to other coffee cultivars, environmental circumstances, and leaf developmental phases. Predictive power could be further increased by including multispectral imagery or temporal growth data. Overall, the findings show that XViT is an effective and comprehensible framework for managing the nutrients in coffee leaves, offering practical insights and a standard for upcoming models in precision agriculture.

Key discussion points:

- The proposed XViT model outperformed ResNet-50, EfficientNet-B4, and ConvNeXt in nutrient deficiency classification under identical experimental settings, achieving superior precision, recall, F1-score, and accuracy.
- The framework integrates XViT-based nutrient deficiency detection with a transparent rule-based treatment recommendation module, ensuring

interpretable and expert-guided treatment suggestions without treating treatment recommendation as an independent learning task.

- The Vision Transformer architecture effectively captures global contextual information and long-range dependencies, improving discrimination of visually similar nutrient deficiency symptoms compared with CNN-based models.
- Data augmentation enhanced model robustness by improving training diversity and recognition of minority nutrient deficiency classes.
- Grad-CAM and attention visualization improve model transparency by highlighting image regions influencing predictions, supporting user trust and practical agricultural deployment.
- The rule-based recommendation module provides consistent and agronomical meaningful treatment suggestions through an expert-defined knowledge base, achieving complete rule coverage and strong expert agreement.
- The proposed framework demonstrates potential for precision coffee nutrient management by combining accurate diagnosis with interpretable decision support.
- Future work will explore severity-aware recommendations, soil and environmental factors, multispectral imaging, temporal monitoring, diverse coffee cultivars, and field-scale validation.

6.1 Dataset size and overfitting considerations

Although the proposed model performs well, the dataset employed in this work is quite limited, with just 1,196 photos compared to larger benchmarks. Such minimal data may raise the likelihood of overfitting, in which the model learns dataset-specific patterns rather than generalizable characteristics.

To address this issue, numerous solutions were implemented. First, substantial data augmentation techniques such as rotation, flipping, scaling, and brightness manipulation were used to artificially increase data diversity. Second, transfer learning was used, which enabled the model to use pretrained weights from large-scale datasets, enhancing generalization. Third, regularization techniques like dropout were used to keep the model from memorizing training data. To ensure robustness, the model's performance was assessed using distinct training and validation splits.

Despite these efforts, the limited dataset size remains a constraint, and the model's generalization capability may be further improved by incorporating larger and more diverse datasets collected under varying environmental conditions. Future work will focus on expanding the dataset and validating the model across different geographical regions.

6.2 Limitations

Despite its high performance, the proposed framework has a few restrictions. The dataset's quantity and diversity are restricted, which may impact generalization across different coffee types, geographies, and environmental conditions. The current model is based mostly on leaf images and does not take into account additional parameters such as soil conditions, climate, or plant growth stage, all of which influence nutrient deficits. Furthermore, the rule-based suggestion module is based on expert-defined knowledge and may require adaptation to diverse agricultural practices. Additional field validation

with larger and more diverse datasets is required to increase robustness and real-world deployment capability.

6.3 Practical implications and deployment

The proposed XViT-based approach has the potential to improve precision coffee growing by delivering quick and interpretable nutrient shortage diagnosis and expert-guided fertilizer recommendations. In real-world applications, farmers or agricultural extension workers can use mobile devices or field cameras to collect photos of coffee leaves, which the model subsequently processes to indicate potential nutrient deficits. The projected deficiency class is then transformed into a suitable fertilizer recommendation using the rule-based knowledge module, allowing for timely and informed nutrient management decisions.

The incorporation of Grad-CAM-based explanations boosts user trust by showing the leaf regions that drive the model's predictions, allowing agronomists to confirm whether the reported symptoms match meaningful deficiency patterns. This combination of automated diagnosis, transparent reasoning, and expert-based recommendations decreases reliance on manual inspection while promoting more efficient crop monitoring.

The proposed framework can be deployed as a lightweight mobile or cloud-based decision assistance application. A mobile-based system would allow for field-level diagnosis in areas with limited access to agricultural experts, whereas a cloud-based platform could provide centralized monitoring, model updates, and integration with additional agricultural data sources such as soil properties, environmental conditions, and farm management records. Future deployment studies should test the model in a variety of field situations, coffee types, and geographical regions to determine its robustness and scalability.

7 Conclusion and future work

7.1 Conclusion

This study proposed an explainable Vision Transformer (XViT)-based framework for automated coffee leaf nutrient deficiency classification and treatment recommendation. The framework integrates class-wise data augmentation, an XViT classifier, Grad-CAM explainability, and a rule-based recommendation module to provide accurate, transparent, and practical decision support for precision coffee farming. Experimental results demonstrated that the proposed framework consistently outperformed ResNet-50, EfficientNet-B4, and ConvNeXt under identical experimental conditions. The proposed XViT achieved a classification accuracy of $94 \pm 0.37\%$ on the original dataset and $96 \pm 0.29\%$ on the augmented dataset, showing that data augmentation effectively improved class balance and enhanced the recognition of minority nutrient deficiency classes. Statistical analysis further confirmed that the performance improvements over the baseline models were statistically significant ($p < 0.001$). In addition, Grad-CAM visualizations demonstrated that the model focused on agronomical meaningful symptom regions, improving the interpretability of its predictions. The rule-based recommendation module also provided consistent and expert-aligned treatment recommendations, achieving 100% rule coverage, 100% rule consistency, and a 93.72% end-to-end recommendation success rate. Overall, the proposed framework offers an

accurate, explainable, and reliable decision-support system with strong potential to support sustainable coffee nutrient management and precision agriculture.

7.2 Future work

Future studies will focus on enhancing the robustness and practical application of the proposed framework. A wider and more diverse coffee leaf dataset collected from different geographic regions and climatic conditions will be used to boost the model's generalization capabilities. Additional nutrient deficiencies, plant diseases, and combined stress conditions can also be incorporated to develop a more comprehensive crop health diagnosis system.

Furthermore, the framework can be expanded to include environmental parameters such as soil nutrient information, meteorological conditions, and IoT sensor data, resulting in more context-aware treatment recommendations. Future research may look into advanced explainable AI techniques, such as SHAP and LIME, to supplement Grad-CAM and improve model transparency. Finally, adopting the proposed framework as a lightweight mobile or cloud-based application would allow for real-time nutrient deficiency diagnosis and treatment recommendations, making it more useful for precision coffee farming and sustainable agricultural management.

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Author contributions

Fikadu Berie Adugna conceptualized and designed the study, developed the Explainable Vision Transformer for Coffee Leaf Nutrient Deficiency Detection with Rule-Based Treatment Recommendation model, conducted data collection and preprocessing, performed experiments and evaluations, implemented explainable AI methods, analyzed and interpreted the results, and drafted, reviewed, and finalized the manuscript. All work reported in this study was carried out solely by the author.

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Data availability

The compiled image dataset used during the current study is available in (<https://www.kaggle.com/datasets/alinedobrovsky/plant-disease-classification-merged-dataset>) and (<https://data.mendeley.com/datasets/brfgw46wzb/1>). The compiled image dataset, source code, trained models, XViT framework, and rule-based treatment recommendation module developed in this study are available from the corresponding author, Fikadu Berie Adugna, upon reasonable request to support reproducibility (fikadu.berie@uog.edu.et).

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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